

PROJECT PROPOSAL & TECHNICAL CHARTER

Getthawha Omnichannel Spa & Wellness Booking Platform

Project Title	Getthawha Omnichannel Spa & Wellness Platform
Document Type	Commercial Proposal & Technical Baseline
Standards Alignment	IEEE Std 1058 / PMBOK 7th Edition Guide
Document Version	Version 2.0 (Enterprise Comprehensive Baseline)
Target Organization	Getthawha Thai Massage & Wellness Group, Chiang Mai
Technical Authority	Enterprise Solutions Architecture Team
Effective Date	September 2026
Classification	Commercial Confidential / Technical Baseline

Document Control & Approvals

This document constitutes a baseline engineering deliverable for the Getthawha Platform. Formal review, authorization, and historical modifications are recorded below:

Formal Review & Acceptance Sign-Off

Role / Representation	Name	Title / Organization	Status	Sign Date
Executive Sponsor	Mawin (Project Owner)	Founder & Managing Director	Approved	September 2026
Technical Authority	Solutions Architect	Chief Software Architect	Approved	September 2026
Store Operations	Operations Director	Head of Branch Operations	Approved	September 2026
Quality Assurance	QA Lead	Principal Test Architect	Approved	September 2026

Document Revision History

Version	Release Date	Author / Role	Summary of Technical Changes
1.0	August 1, 2026	Solutions Architect	Initial feasibility proposal and preliminary architecture
1.5	August 20, 2026	Product Engineering Lead	Added multi-channel scope, pricing catalog, and sprint roadmaps
2.0	September 14, 2026	Chief Software Architect	Comprehensive revision: Work Breakdown Structure, RACI, Financial ROI, and Risk Register

Confidentiality & Proprietary Baseline

The information contained in this document is confidential and proprietary to Getthawha Thai Massage & Wellness Group. It is submitted with the understanding that it will be held in strict confidence and will not be disclosed, duplicated, or used, in whole or in part, for any purpose other than evaluating and operating the Getthawha Omnichannel Platform without prior written consent from the executive board.

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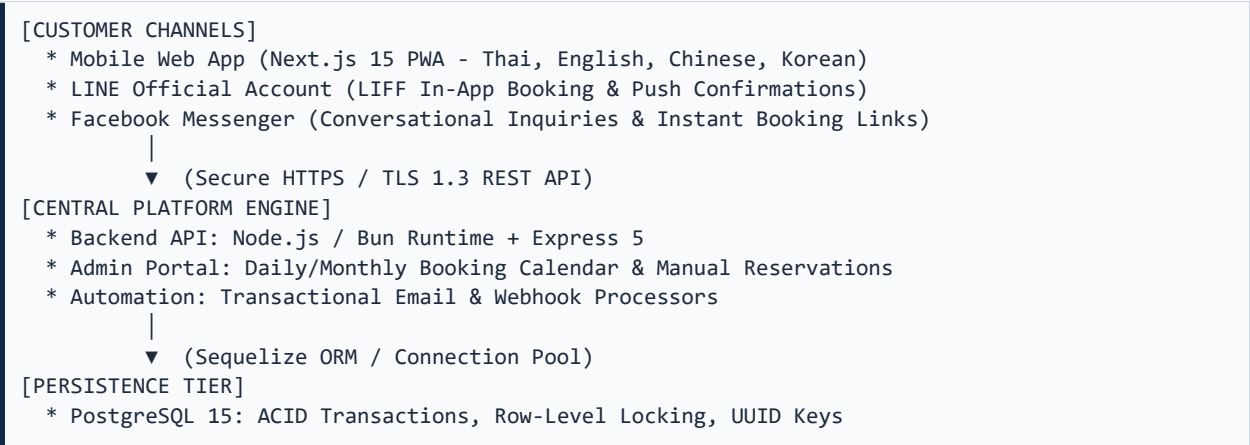
This specification is organized into the following formal sections:

- 1.0 Executive Summary & Platform Overview
- 2.0 Business Case & Problem Statement
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1. Executive Summary

Getthawha Thai Massage & Wellness Group is an established wellness provider operating five flagship branches across Chiang Mai, Thailand. The brand is renowned for preserving traditional Lanna herbal treatments and authentic northern Thai bodywork. Today, Chiang Mai serves as a major international destination for wellness tourism, attracting significant numbers of visitors from English-speaking countries, Greater China, and South Korea, alongside a steady domestic customer base.

To meet modern consumer expectations and optimize front-desk operations, Getthawha is undertaking a full digital transformation. The **Getthawha Omnichannel Booking Platform** replaces fragmented manual scheduling with a unified, cloud-native reservation and store management system.



Core Value Drivers:

- 1. Frictionless Omnichannel Ingress:** Customers can complete reservations via Mobile Web, LINE, or Facebook Messenger within four screen interactions, with no third-party app installations required.
- 2. True Multilingual Experience:** Native interfaces in Thai, English, Simplified Chinese, and Korean with dynamic CJK font loading, eliminating tourist miscommunication.
- 3. Automated Store Discovery:** HTML5 Geolocation computes the customer's distance to all five branches using the Haversine formula, automatically recommending the nearest location.
- 4. Unified Operational Oversight:** Front-desk receptionists and central management work from a synchronized real-time booking calendar, preventing scheduling conflicts and room overbooking.

2. Business Case & Problem Statement

2.1 Current Operational Bottlenecks (As-Is State)

- **Excessive Front-Desk Workload:** Reception staff spend approximately 35% of their daily working hours responding to repetitive inquiries via telephone, LINE chats, and Facebook messages regarding service menus, pricing, branch locations, and available timeslots.
- **Double-Booking and Concurrency Failures:** During peak evening hours (17:00 to 21:00) and tourist seasons, manual paper and spreadsheet scheduling frequently causes overlapping room assignments and unallocated therapists.
- **International Customer Churn:** International tourists encounter language barriers during telephone inquiries and often abandon booking attempts when menus are not clearly explained in their language.
- **Absence of Centralized Data Analytics:** Store managers lack visibility into customer retention, branch-by-branch revenue, top-performing treatments, and voucher marketing attribution.

2.2 Future Operational Objectives (To-Be State)

- **Autonomous Self-Service:** At least 75% of reservations completed directly by customers online.
- **Guaranteed Scheduling Integrity:** Zero double-bookings achieved through database transactions and concurrency control.
- **Immediate Multilingual Onboarding:** Foreign tourists complete reservations in their native language in under two minutes.
- **Automated Customer Retention:** Automated email confirmations and post-visit review requests drive repeat visits and customer loyalty.

3. Strategic Objectives & Success Metrics (KPIs)

Project progress will be tracked and evaluated against the following performance indicators over the initial six-month rollout:

Business Objective	Key Performance Indicator (KPI)	Current Baseline	Target Milestone	Review Cadence
Self-Service Adoption	Share of total bookings made autonomously online	0% (All manual)	≥ 75%	Monthly
Reception Handling Time	Average staff time spent scheduling per reservation	8.5 minutes	≤ 2.0 minutes	Weekly sample
International Bookings	Bookings completed in English, Chinese, or Korean	~15% (Phone)	≥ 45%	Monthly
Scheduling Accuracy	Duplicate room or therapist booking collisions	12 - 15 / month	0 incidents	Continuous
Customer Feedback	Verified reviews collected from customers	< 20 / month	≥ 250 / month	Monthly
System Availability	Core REST API uptime during business hours	N/A	≥ 99.8%	Continuous

4. Scope Baseline: In-Scope vs. Out-of-Scope

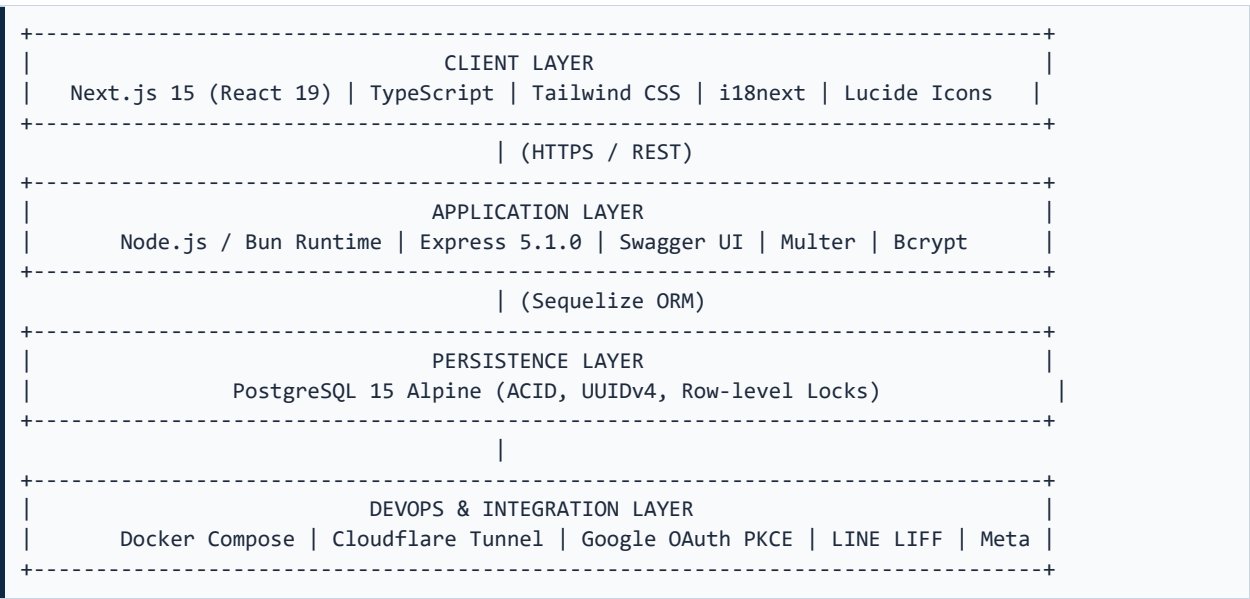
4.1 In-Scope Deliverables

- **Customer Facing Web Application (`getthawha-client-frontend`):**
- Modern, responsive interface tailored for smartphones and tablets.
- Native quadrilingual support: Thai, English, Simplified Chinese, Korean.
- Interactive treatment catalog with dynamic duration toggling (60, 90, 120 minutes) and real-time price recalculation.
- Multi-step reservation wizard (Branch, Package, Date/Timeslot 09:30–20:00, Customer Details, Voucher Code).
- Customer account portal with appointment history and self-service cancellation.
- Social Single Sign-On (SSO) via LINE Login and Google OAuth 2.0 with PKCE security.
- Geolocation-powered nearest branch calculation and Google Maps navigation.
- **Administrative Operations Portal (`getthawha-admin-frontend`):**
- Role-based staff authentication (Bcrypt password hashing + JWT session tokens).
- Master daily schedule grid and monthly booking density calendar.
- Front-desk manual booking modal for walk-in and telephone guests.
- Complete management (CRUD) for Branches, Packages, Promotions, Vouchers, and Staff.
- Customer review moderation interface (Approve / Reject).
- **Backend API & Service Engine (`getthawha-backend`):**
- 26+ RESTful endpoints documented interactively via OpenAPI / Swagger UI.
- Transactional email delivery via Nodemailer SMTP gateway.
- Meta Facebook Messenger Webhook engine with automated quick-reply responses.
- Secure multipart file upload service (Multer) for promotional media.
- **Containerized Infrastructure & Deployment:**
- Docker Compose configurations for development (hot-reload enabled) and production.
- Automated Cloudflare Ingress Tunnel ([cloudflared](#)) for secure external webhook delivery.

4.2 Out-of-Scope (Deferred to Future Releases)

- In-app credit card payment processing (current phase implements reserve-online, pay-at-store).
- Native iOS/Android store application packages (fully served via responsive PWA and LINE LIFF).
- Therapist commission and store payroll accounting.
- Physical warehouse inventory tracking for massage oils and retail supplies.

5. Technology Stack & Technical Justification



Layer / Component	Technology Selected	Evaluation Summary & Engineering Justification
Frontend Framework	Next.js 15 (App Router)	Server-Side Rendering (SSR) delivers superior search engine optimization (SEO) for tourists and rapid initial page loads on mobile networks.
Styling Engine	Tailwind CSS v4	Utility-first architecture ensures zero runtime CSS overhead, high responsiveness across devices, and clean component isolation.
Backend Runtime	Bun / Node.js	Bun provides near-instant container startup and high-speed test suite execution while maintaining complete Express 5 compatibility.
API Framework	Express 5.1.0	Mature, lightweight framework with native Promise error handling and modular routing for microservice endpoints.
Database System	PostgreSQL 15	Guarantees ACID transactional compliance, foreign key integrity, and row-level locking necessary to prevent appointment collisions.
DevOps Ingress	Cloudflare Tunnel	Provides an encrypted ingress route directly to Docker containers without exposing static IP addresses or requiring router port forwarding.

6. Work Breakdown Structure (WBS)

The project is structured into four levels of deliverables to ensure systematic execution and accountability:

- **1.0 Getthawha Platform Delivery**
- **1.1 Core Architecture & Persistence (Sprint 1)**
 - 1.1.1 Database Schema Modeling & Sequelizeize Entities (**User, Branch, Package, Booking, Voucher, Review**)
 - 1.1.2 Docker Multi-Container Network Configuration (**docker-compose.dev.yml, docker-compose.yml**)
 - 1.1.3 Centralized Express 5 Routing & Middleware Pipeline
 - 1.1.4 Interactive Swagger / OpenAPI 3.0 Documentation Setup
- **1.2 Administrative Operations Portal (Sprint 1 - 2)**
 - 1.2.1 Secure Staff Authentication (Bcrypt + JWT + HttpOnly Cookies)
 - 1.2.2 Master Booking Calendar (Monthly aggregation and daily timeslot schedule)
 - 1.2.3 Manual Walk-in & Phone Reservation Interface
 - 1.2.4 Master Data Management (Branches, Packages, Promotions, Vouchers)
 - 1.2.5 Customer Review Moderation Workflow
- **1.3 Customer Experience & Booking Wizard (Sprint 2)**
 - 1.3.1 Mobile-First UI/UX Built to Figma Specifications
 - 1.3.2 Quadrilingual Localization Engine (TH, EN, ZH, KO) & Dynamic Typography
 - 1.3.3 Geolocation Proximity Engine with Haversine Nearest Branch Calculation
 - 1.3.4 Dynamic Duration Selector (60m/90m/120m) & Real-Time Price Calculation
 - 1.3.5 Voucher Verification & Server-Side Discount Calculation
 - 1.3.6 Customer Profile, Booking History & Self-Cancellation Flow
- **1.4 Omnichannel Integrations & Social Commerce (Sprint 2)**
 - 1.4.1 Google OAuth 2.0 Single Sign-On with PKCE Code Challenge
 - 1.4.2 LINE Official Account Integration & LIFF In-App Webview
 - 1.4.3 Meta Facebook Messenger Webhook Handshake & Conversational Bot
 - 1.4.4 Transactional Email Notification Dispatcher via Nodemailer
- **1.5 Quality Assurance, Security & Production Deployment (Sprint 3)**
 - 1.5.1 Automated Unit & Integration Testing (**bun test**)
 - 1.5.2 Facebook Bot Handshake Verification Script (**check-facebook-bot.ps1**)
 - 1.5.3 Cloud VPS Production Deployment, Reverse Proxy & SSL Configuration
 - 1.5.4 Disaster Recovery Drills & Automated Database Backup Routines

7. RACI Responsibility Assignment Matrix

Roles correspond to the project governance model established in the Getthawha Project Hub:

- **Roles:**
- **PL:** Project Lead & Fullstack Solutions Architect
- **FE:** Frontend Applications Engineer
- **BE:** Backend API & Integrations Engineer
- **QA:** Quality Assurance & Product Specialist
- **Key:**
- **R (Responsible):** The role directly performing the deliverable.
- **A (Accountable):** The individual with final approval authority.
- **C (Consulted):** Key stakeholder providing technical or domain input.
- **I (Informed):** Individuals kept updated on progress.

WBS Deliverable / Activity	Project Lead (PL)	Frontend Dev (FE)	Backend Dev (BE)	QA / Product (QA)
System Architecture & Database Modeling	A / R	C	R	I
Docker Multi-Container Orchestration	A / R	I	R	I
Client Frontend UI & Mobile Responsiveness	A	R	C	C
Quadrilingual Localization (TH/EN/ZH/KO)	A	R	I	C
Admin Booking Calendar & Walk-in Modal	A	R	C	C
Express RESTful Endpoints & OpenAPI Spec	A	I	R	C
Google OAuth 2.0 PKCE Implementation	A	C	R	I
Meta Facebook Messenger Webhook & Bot	A	I	R	C
LINE LIFF In-App Integration	A	C	R	C
Automated Test Suites & Regression Runs	A	C	C	R
Disaster Recovery Drills & Backup Verification	A / R	I	R	I
Final User Acceptance Testing (UAT)	A	C	C	R

8. Milestone Schedule & Delivery Roadmaps

Sprint 1: Core Foundation (Weeks 1 - 3)

- [X] Database Schema & Sequelize Relational Models
- [X] Express 5 Backend REST Engine & Swagger UI
- [X] Admin Operations Portal CRUD & Calendar Roster

Sprint 2: Omnichannel Experience (Weeks 4 - 6)

- [X] Customer Web Booking Wizard & Dynamic Duration Engine
- [X] Google OAuth 2.0 PKCE & LINE LIFF In-App Flow
- [X] Meta Facebook Messenger Bot Webhook & Cloudflare Tunnel
- [X] Quadrilingual Localization (TH / EN / ZH / KO) & CJK Fonts

Sprint 3: Production Hardening (Weeks 7 - 8)

- [>] Automated Testing & Regression Test Harness
- [] Cloud VPS Production Launch & Nginx Reverse Proxy
- [] Final UAT Sign-Off & Store Reception Training

9. Risk Management Register

Risk ID	Risk Event & Potential Impact	Likelihood	Impact	Preventative Mitigation Strategy	Contingency Plan
RSK-01	OAuth Callback Mismatch: Google redirects fail on production due to protocol, port, or trailing slash discrepancies.	Medium	High	Maintain strict environment parity (<code>GOOGLE_REDIRECT_URI</code>) across dev and prod. Unit tests in <code>google.config.test.js</code> .	Revert to fallback LINE Login or direct booking workflow.
RSK-02	Booking Concurrency Collision: Multiple customers reserve the same room or therapist simultaneously.	Medium	High	Atomic PostgreSQL transactions with capacity checks in <code>booking.validation.js</code> .	Prompt customer with a clear 409 Conflict message and suggest adjacent timeslots.
RSK-03	Meta Facebook API Version Drift: Meta updates Graph API parameters, breaking webhook responses.	Medium	Medium	Pin API version to <code>v21.0</code> ; verify handshake automatically using <code>check-facebook-bot.ps1</code> .	Deploy updated webhook route container using Docker hot-reload.
RSK-04	Data Privacy Violation (PDPA/GDPR): Unauthorized exposure of customer phone numbers or email addresses.	Low	Critical	Store session JWTs in <code>HttpOnly</code> Secure cookies; exclude user contact data from public API endpoints.	Immediate incident disclosure protocol and session token invalidation.
RSK-05	Next.js Hydration Mismatch: Timestamp discrepancies between server UTC clock and customer browser local time.	Medium	Low	Wrap date renders in client-mounted hooks and set container timezone explicitly to <code>Asia/Bangkok</code> .	Add <code>suppressHydrationWarning</code> on date-rendering components.

10. Financial Analysis: Return on Investment (ROI) & TCO

10.1 Operational Cost Savings (Monthly Estimate)

- **Front-Desk Hours Saved:** 5 branches × 2 staff × 2 hours/day × 30 days = **600 hours / month**.
- **Direct Labor Savings:** 600 hours × 100 THB/hour = **60,000 THB / month**.
- **Recovered International Bookings:** Estimated 45 additional bookings/month × 1,200 THB average ticket = **54,000 THB / month**.
- **Total Estimated Value Generated:** **~114,000 THB / month**.

10.2 Total Cost of Ownership (Cloud VPS Infrastructure)

- Cloud VPS Host (Ubuntu 22.04 LTS, 4 vCPU, 8GB RAM): ~850 THB / month (\$24).
- Domain & Managed DNS (Cloudflare): ~35 THB / month (\$12 / year).
- Transactional Email Gateway (SMTP): Free tier up to 10,000 emails.
- **Total Monthly Infrastructure Overhead:** **< 1,000 THB / month**.
- **Project Payback Period:** Estimated at **less than 1.5 months** following production go-live.

11. Service Level Agreement (SLA) & Operational Commitments

- **Service Availability:** The platform targets $\geq 99.8\%$ **operational uptime** during business operating hours (08:00 to 22:00 Asia/Bangkok time).
- **Recovery Time Objective (RTO):** In the event of primary server failure, system operation will be restored from backup within **< 30 minutes**.
- **Recovery Point Objective (RPO):** Maximum allowable data loss in the event of hardware failure is **< 24 hours** (governed by daily automated backup routines).
- **Security Auditing:** Docker base images and application dependencies will undergo automated monthly vulnerability scans.

12. Stakeholder Approval & Formal Sign-Off

By signing below, the authorized representatives acknowledge that this Project Proposal accurately represents the agreed scope, technical architecture, and implementation roadmap:

Representative Role	Full Name	Signature	Approval Date
Executive Sponsor	Management Representative	_____	___ / ___ / 2026
Project Lead & Architect	Mawin Skalet	_____	___ / ___ / 2026
Lead Solutions Engineer	Fullstack Engineering Lead	_____	___ / ___ / 2026
Quality Assurance Lead	QA & Product Specialist	_____	___ / ___ / 2026